



Solar Roadmap for Assam

Tariff Analysis for the 1000 MW Solar Project

September 2023

Assumptions for Base Case : 1000 MW Solar Plant with Grant (80% of Project Cost)

S.N	Parameters	Units	Input	Comments
A.	Project Cost Input			
	Project Size-Contracted Capacity(AC)	MW	750	As per PFS Report
	Project Size -Installed Capacity(DC)	MWp	1,000	As per PFS Report
	PPA Effective Start Date	Date	1-Apr-24	Assumption
	Overloading Factor		1.3	As per PFS Report
	Total Project Cost /MW	INR Cr/MW	4.71	As per PFS Report + IDC
	Total Project Cost	INR Cr	3,533	
	Construction Start Date	Date	1-Apr-24	Assumption
	Construction period	Years	2.0	Assumption
	Project COD	Date	1-Apr-26	Assumption
	Project Life	Years	25	As per Norms
B.	Finance Cost			
	Grant:Debt:Equity Ratio	x:1 ratio	80:0:20	As per Project Plan
	Interest cost till COD	%	9.50%	Assumption; As per AERC it would be 10.50%
	Repayment period	years	13	As per AERC norms
	Moratorium Period (post COD)	years	1	Assumption
	Upfront Equity Requirement as % of total equity	%	30%	Assumption
	Cost of Equity	%	14.0%	As per AERC norms
	WACC	%	14.00%	As per AERC norms
	Capex Phasing	%	FY25,26:50%	Assumption
C.	Capacity Utilization			
	CUF - first year (Based on Contracted Capacity)	%	20.98%	As per PFS Report
	Auxiliary & Transmisison loss	%	0.0%	
	Degradation Annually	%	0.51%	Assumption

S.N	Parameters	Units	Input	Comments
D.	O & M Cost			
	Site O&M expenses	INR lakh/mw	6.00	As per PFS Report
	Site O&M Expenses escalation	%	3.84%	As per PFS Report
E.	Working Capital Requirement			
	Receivables - tariff	Months	2.0	As per AERC norms
	O&M Expenses	Months	1	As per AERC norms
	Maintenance spares	of O & M	15%	As per AERC norms
	WC Interest cost	%	10.50%	Assumption; As per AERC it would be 11.50%
F.	Taxation			
	Income tax rate	%	29.12%	As per prevailing taxation rules
	MAT rate	%	17.47%	
	MAT lapsation period	years	15	
	Average tax over the life of the plant	%	23.25%	
	Benefit of Section 80 IA of IT Act		Yes	
	Income tax rate - New Regime / Concessional	%	25.17%	
G.	Depreciation			
	Max permissible depreciation	%	90%	As per AERC norms
	Grant as percentage of project cost	%	80.00%	As per Project Plan
	Book Depreciation Rate - SLM	%	5.28%	As per AERC norms

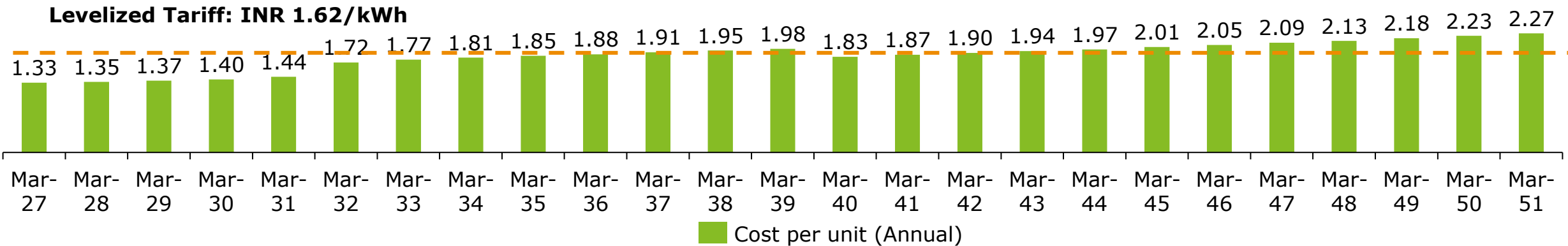
Assumptions for Scenario 1: 1000 MW Solar Plant without Grant (70:30 Debt/Equity)

S.N	Parameters	Units	Input	Comments
A.	Project Cost Input			
	Project Size-Contracted Capacity(AC)	MW	750	As per PFS Report
	Project Size -Installed Capacity(DC)	MWp	1,000	As per PFS Report
	PPA Effective Start Date	Date	1-Apr-24	Assumption
	Overloading Factor		1.3	As per PFS Report
	Total Project Cost /MW	INR Cr/MW	4.71	As per PFS Report + IDC
	Total Project Cost	INR Cr	3,533	
	Construction Start Date	Date	1-Apr-24	Assumption
	Construction period	Years	2.0	Assumption
	Project COD	Date	1-Apr-26	Assumption
	Project Life	Years	25	As per Norms
B.	Finance Cost			
	Grant:Debt:Equity Ratio	x:1 ratio	0:70:30	As per Project Plan
	Interest cost till COD	%	9.50%	Assumption; As per AERC it would be 10.50%
	Repayment period	years	13	As per AERC norms
	Moratorium Period (post COD)	years	1	Assumption
	Upfront Equity Requirement as % of total equity	%	30%	Assumption
	Cost of Equity	%	14.0%	As per AERC norms
	WACC	%	9.54%	As per AERC norms
	Capex Phasing	%	FY25,26:50%	Assumption
C.	Capacity Utilization			
	CUF - first year (Based on Contracted Capacity)	%	20.98%	As per PFS Report
	Auxiliary & Transmisison loss	%	0.0%	
	Degradation Annually	%	0.51%	Assumption

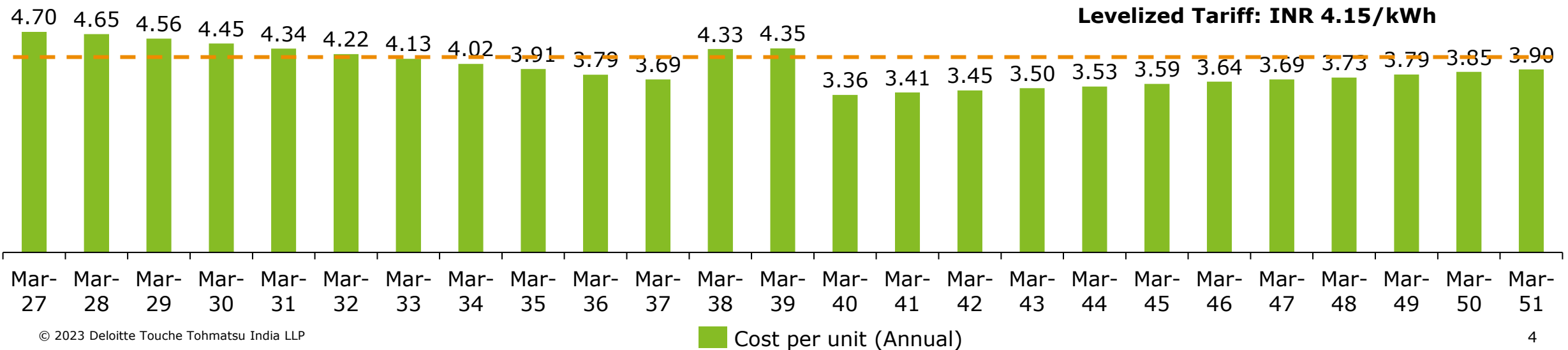
S.N	Parameters	Units	Input	Comments
D.	O & M Cost			
	Site O&M expenses	INR lakh/mw	6.00	As per PFS Report
	Site O&M Expenses escalation	%	3.84%	As per PFS Report
E.	Working Capital Requirement			
	Receivables - tariff	Months	2.0	As per AERC norms
	O&M Expenses	Months	1	As per AERC norms
	Maintenance spares	of O & M	15%	As per AERC norms
	WC Interest cost	%	10.50%	Assumption; As per AERC it would be 11.50%
F.	Taxation			
	Income tax rate	%	29.12%	As per prevailing taxation rules
	MAT rate	%	17.47%	
	MAT lapsation period	years	15	
	Average tax over the life of the plant	%	23.25%	
	Benefit of Section 80 IA of IT Act		Yes	
	Income tax rate - New Regime / Concessional	%	25.17%	
G.	Depreciation			
	Max permissible depreciation	%	90%	As per AERC norms
	Grant as percentage of project cost	%	80.00%	As per Project Plan
	Book Depreciation Rate - SLM	%	5.28%	As per AERC norms

Model Outputs – Levelized and YoY Tariffs

Base Case: 1000 MW Solar Plant with Grant (80% of Project Cost)



Scenario 1: 1000 MW Solar Plant without Grant (70:30 Debt/Equity)





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